

数学笔记

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Chapter 1

Symplectic Geometry

1.1 Hamiltonian Mechanics

Hamilton's Equations

Hamiltonian mechanics is a reformulation of classical mechanics. It arises from Lagrangian mechanics by a Legendre transformation. The state of a mechanical system is described by a point in a $2n$ -dimensional phase space, with coordinates $(q_1, \dots, q_n, p_1, \dots, p_n)$, where the q_i are the generalized coordinates, and the p_i are the conjugate momenta. The time evolution of the system is given by **Hamilton's equations**:

$$\begin{cases} \frac{\partial q_i}{\partial t} = \frac{\partial H}{\partial p_i}, \\ \frac{\partial p_i}{\partial t} = -\frac{\partial H}{\partial q_i}, \end{cases} \quad i = 1, \dots, n,$$

where $H(q, p, t)$ is the **Hamiltonian function**, which is usually the total energy of the system.

Suppose $(q(t), p(t))$ is a solution to Hamilton's equations. Then for any function $f(q, p, t)$, its time evolution is given by

$$\frac{df}{dt} = \frac{\partial f}{\partial t} + \sum_{i=1}^n \left(\frac{\partial f}{\partial q_i} \frac{dq_i}{dt} + \frac{\partial f}{\partial p_i} \frac{dp_i}{dt} \right) = \frac{\partial f}{\partial t} + \sum_{i=1}^n \left(\frac{\partial f}{\partial q_i} \frac{\partial H}{\partial p_i} - \frac{\partial f}{\partial p_i} \frac{\partial H}{\partial q_i} \right).$$

We define the **Poisson bracket** of two functions f and g by

$$\{f, g\} = \sum_{i=1}^n \left(\frac{\partial f}{\partial q_i} \frac{\partial g}{\partial p_i} - \frac{\partial f}{\partial p_i} \frac{\partial g}{\partial q_i} \right).$$

Then the time evolution of f can be written as

$$\frac{df}{dt} = \frac{\partial f}{\partial t} + \{f, H\}.$$

So f is a constant of motion if and only if $\frac{\partial f}{\partial t} + \{f, H\} = 0$. If in particular f does not depend on t , then f is a constant of motion if and only if $\{f, H\} = 0$.

Hamilton's Equations and Symplectic Forms

To formulate Hamiltonian mechanics in a symplectic geometric framework, we introduce the symplectic form ω on the phase space:

$$\omega = \sum_{i=1}^n dq_i \wedge dp_i.$$

This is a closed, non-degenerate 2-form. Let $H = H(q, p)$ be a Hamiltonian function. The Hamiltonian vector field X_H associated to H is defined by the equation

$$\iota_{X_H} \omega = dH,$$

where ι_{X_H} denotes the interior product with X_H . The left-hand side is

$$\iota_{X_H} \omega = \sum_{i=1}^n \left(dq_i(X_H) dp_i - dp_i(X_H) dq_i \right).$$

The right-hand side is

$$dH = \sum_{i=1}^n \left(\frac{\partial H}{\partial q_i} dq_i + \frac{\partial H}{\partial p_i} dp_i \right).$$

So we have

$$dq_i(X_H) = \frac{\partial H}{\partial p_i}, \quad dp_i(X_H) = -\frac{\partial H}{\partial q_i}.$$

Thus the Hamiltonian vector field is

$$X_H = \sum_{i=1}^n \left(\frac{\partial H}{\partial p_i} \frac{\partial}{\partial q_i} - \frac{\partial H}{\partial q_i} \frac{\partial}{\partial p_i} \right).$$

The time evolution of the system is given by the flow of X_H , which is exactly Hamilton's equations.

For any smooth functions f , we can uniquely determine a vector field X_f by solving the equation

$$\iota_{X_f} \omega = df.$$

For any two smooth functions f, g , We can define the Poisson bracket to be

$$\{f, g\} := \omega(X_f, X_g).$$

When $\omega = \sum_{i=1}^n dq_i \wedge dp_i$, we can get

$$X_f = \sum_{i=1}^n \left(\frac{\partial f}{\partial p_i} \frac{\partial}{\partial q_i} - \frac{\partial f}{\partial q_i} \frac{\partial}{\partial p_i} \right).$$

So

$$\begin{aligned} \omega(X_f, X_g) &= \sum_{i,j=1}^n \omega \left(\frac{\partial f}{\partial p_i} \frac{\partial}{\partial q_i} - \frac{\partial f}{\partial q_i} \frac{\partial}{\partial p_i}, \frac{\partial g}{\partial p_j} \frac{\partial}{\partial q_j} - \frac{\partial g}{\partial q_j} \frac{\partial}{\partial p_j} \right) \\ &= \sum_{i,j=1}^n \left(\frac{\partial f}{\partial q_i} \frac{\partial g}{\partial p_j} - \frac{\partial f}{\partial p_i} \frac{\partial g}{\partial q_j} \right) \delta_{ij} \\ &= \sum_{i=1}^n \left(\frac{\partial f}{\partial q_i} \frac{\partial g}{\partial p_i} - \frac{\partial f}{\partial p_i} \frac{\partial g}{\partial q_i} \right). \end{aligned}$$

Canonical Transformations

Let $(Q(q, p), P(q, p))$ be a transformation from (q, p) to (Q, P) . We say it is a **canonical transformation** if it preserves the form of Hamilton's equations, i.e., if $(Q(t), P(t))$ satisfies

$$\begin{cases} \frac{\partial Q_i}{\partial t} = \frac{\partial K}{\partial P_i}, \\ \frac{\partial P_i}{\partial t} = -\frac{\partial K}{\partial Q_i}, \end{cases} \quad i = 1, \dots, n,$$

for some Hamiltonian function $K(Q, P, t)$. This is equivalent to the **Poisson bracket condition**:

$$\begin{aligned} \{P_i, Q_j\}_{q,p} &= \delta_{ij}, \\ \{Q_i, Q_j\}_{q,p} &= 0, \\ \{P_i, P_j\}_{q,p} &= 0. \end{aligned}$$

Another equivalent condition is the **differential form condition**:

$$\sum_{i=1}^n dQ_i \wedge dP_i = \sum_{i=1}^n dq_i \wedge dp_i.$$

Canonical transformations and Symplectic Forms

Let $(Q(q, p), P(q, p))$ be a transformation from (q, p) to (Q, P) . The Jacobian matrix of this transformation is

$$J = \frac{\partial(Q, P)}{\partial(q, p)} = \begin{pmatrix} \frac{\partial Q}{\partial q} & \frac{\partial Q}{\partial p} \\ \frac{\partial P}{\partial q} & \frac{\partial P}{\partial p} \end{pmatrix} =: \begin{pmatrix} A & B \\ C & D \end{pmatrix}.$$

Theorem 1.1.1. *The Poisson bracket condition is equivalent to J^T being symplectic.*

Proof. The Poisson bracket condition is

$$\begin{aligned} \{P_k, Q_l\}_{q,p} &= \sum_{i=1}^n \left(\frac{\partial Q_l}{\partial q_i} \frac{\partial P_k}{\partial p_i} - \frac{\partial Q_l}{\partial p_i} \frac{\partial P_k}{\partial q_i} \right) = \delta_{kl}, \\ \{Q_k, Q_l\}_{q,p} &= \sum_{i=1}^n \left(\frac{\partial Q_k}{\partial q_i} \frac{\partial Q_l}{\partial p_i} - \frac{\partial Q_k}{\partial p_i} \frac{\partial Q_l}{\partial q_i} \right) = 0, \\ \{P_k, P_l\}_{q,p} &= \sum_{i=1}^n \left(\frac{\partial P_k}{\partial q_i} \frac{\partial P_l}{\partial p_i} - \frac{\partial P_k}{\partial p_i} \frac{\partial P_l}{\partial q_i} \right) = 0. \end{aligned}$$

Written in a tigher form, they are

$$\begin{aligned} \frac{\partial Q}{\partial q} \left(\frac{\partial P}{\partial p} \right)^T - \frac{\partial Q}{\partial p} \left(\frac{\partial P}{\partial q} \right)^T &= I \Rightarrow AD^T - BC^T = I, \\ \frac{\partial Q}{\partial q} \left(\frac{\partial Q}{\partial p} \right)^T - \frac{\partial Q}{\partial p} \left(\frac{\partial Q}{\partial q} \right)^T &= 0 \Rightarrow AB^T - BA^T = 0, \\ \frac{\partial P}{\partial q} \left(\frac{\partial P}{\partial p} \right)^T - \frac{\partial P}{\partial p} \left(\frac{\partial P}{\partial q} \right)^T &= 0 \Rightarrow CD^T - DC^T = 0. \end{aligned}$$

This is equivalent to say

$$\begin{aligned} J\Omega J^T &= \begin{pmatrix} A & B \\ C & D \end{pmatrix} \begin{pmatrix} 0 & I \\ -I & 0 \end{pmatrix} \begin{pmatrix} A^T & C^T \\ B^T & D^T \end{pmatrix} \\ &= \begin{pmatrix} AB^T - BA^T & AD^T - BC^T \\ CB^T - DA^T & CD^T - DC^T \end{pmatrix} \\ &= \begin{pmatrix} 0 & I \\ -I & 0 \end{pmatrix} = \Omega. \end{aligned}$$

□

Theorem 1.1.2. *The differential form condition is equivalent to J being symplectic.*

Proof. Let $\omega_{(q,p)} = \sum_{i=1}^n dq_i \wedge dp_i$, $\omega_{(Q,P)} = \sum_{i=1}^n dQ_i \wedge dP_i$. Then

$$\begin{aligned} \omega_{(Q,P)} &= \sum_{i=1}^n \left(\sum_{k=1}^n \frac{\partial Q_i}{\partial q_k} dq_k + \frac{\partial Q_i}{\partial p_k} dp_k \right) \wedge \left(\sum_{l=1}^n \frac{\partial P_i}{\partial q_l} dq_l + \frac{\partial P_i}{\partial p_l} dp_l \right) \\ &= \sum_{i=1}^n \sum_{k,l=1}^n \left(\frac{\partial Q_i}{\partial q_k} \frac{\partial P_i}{\partial p_l} - \frac{\partial Q_i}{\partial p_l} \frac{\partial P_i}{\partial q_k} \right) dq_k \wedge dp_l \\ &\quad + \sum_{i=1}^n \sum_{k < l} \left(\frac{\partial Q_i}{\partial q_k} \frac{\partial P_i}{\partial q_l} - \frac{\partial Q_i}{\partial q_l} \frac{\partial P_i}{\partial q_k} \right) dq_k \wedge dq_l \\ &\quad + \sum_{i=1}^n \sum_{k < l} \left(\frac{\partial Q_i}{\partial p_k} \frac{\partial P_i}{\partial p_l} - \frac{\partial Q_i}{\partial p_l} \frac{\partial P_i}{\partial p_k} \right) dp_k \wedge dp_l. \end{aligned}$$

Thus $\omega_{(Q,P)} = \omega_{(q,p)}$ if and only if

$$\begin{aligned} \sum_{i=1}^n \left(\frac{\partial Q_i}{\partial q_k} \frac{\partial P_i}{\partial p_l} - \frac{\partial Q_i}{\partial p_l} \frac{\partial P_i}{\partial q_k} \right) &= \delta_{kl}, \\ \sum_{i=1}^n \left(\frac{\partial Q_i}{\partial q_k} \frac{\partial P_i}{\partial q_l} - \frac{\partial Q_i}{\partial q_l} \frac{\partial P_i}{\partial q_k} \right) &= 0, \\ \sum_{i=1}^n \left(\frac{\partial Q_i}{\partial p_k} \frac{\partial P_i}{\partial p_l} - \frac{\partial Q_i}{\partial p_l} \frac{\partial P_i}{\partial p_k} \right) &= 0. \end{aligned}$$

Written in a tigher form, they are

$$\begin{aligned} \left(\frac{\partial Q}{\partial q} \right)^T \frac{\partial P}{\partial p} - \left(\frac{\partial P}{\partial q} \right)^T \frac{\partial Q}{\partial p} &= I \Rightarrow A^T D - C^T B = I, \\ \left(\frac{\partial Q}{\partial q} \right)^T \frac{\partial Q}{\partial p} - \left(\frac{\partial P}{\partial q} \right)^T \frac{\partial P}{\partial p} &= 0 \Rightarrow A^T B - C^T A = 0, \\ \left(\frac{\partial Q}{\partial p} \right)^T \frac{\partial P}{\partial q} - \left(\frac{\partial P}{\partial p} \right)^T \frac{\partial Q}{\partial q} &= 0 \Rightarrow B^T D - D^T B = 0. \end{aligned}$$

This is equivalent to say

$$\begin{aligned} J^T \Omega J &= \begin{pmatrix} A^T & C^T \\ B^T & D^T \end{pmatrix} \begin{pmatrix} 0 & I \\ -I & 0 \end{pmatrix} \begin{pmatrix} A & B \\ C & D \end{pmatrix} \\ &= \begin{pmatrix} A^T C - C^T A & A^T D - C^T B \\ B^T C - D^T A & B^T D - D^T B \end{pmatrix} \\ &= \begin{pmatrix} 0 & I \\ -I & 0 \end{pmatrix} = \Omega. \end{aligned}$$

□

Theorem 1.1.3. *J is symplectic if and only if J^T is symplectic.*

Proof.

$$\begin{aligned}
 J \text{ is symplectic} &\Leftrightarrow J^T \Omega J = \Omega \\
 &\Leftrightarrow J^{-1} \Omega (J^T)^{-1} = \Omega \\
 &\Leftrightarrow J \Omega J^T = \Omega \\
 &\Leftrightarrow J^T \text{ is symplectic.}
 \end{aligned}$$

□

Those three theorems together give the following conclusion:

Theorem 1.1.4. *The Poisson bracket condition is equivalent to the differential form condition.*